Question 2 (Combined Operators & Superposition)

Problem Statement

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4 \tan^{-1}\left(\frac{y}{x}\right)$, find the value of:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \quad \text{at } x = 1, y = 1$$

Step-by-Step Solution

Step 1: Split the Function Using Superposition

The given function contains two distinct polynomial degrees acting on the trigonometric wrappers. Attempting to determine a single degree of homogeneity for the entire expression directly will fail. Instead, break u down into two independent component functions:

$$u = v + w$$

Where:

$$v = x^3 \sin^{-1}\left(\frac{y}{x}\right)$$

$$w = x^4 \tan^{-1}\left(\frac{y}{x}\right)$$

Step 2: Determine the Degrees of v and w

For Function v:

Substitute $x \to xt$ and $y \to yt$:

$$v(xt, yt) = (xt)^3 \sin^{-1}\left(\frac{yt}{xt}\right)$$

$$v(xt, yt) = t^3 \cdot x^3 \sin^{-1}\left(\frac{y}{x}\right) = t^3 v(x, y)$$

Thus, v is a homogeneous function of degree $n_1 = 3$.

For Function w:

Substitute $x \to xt$ and $y \to yt$:

$$w(xt, yt) = (xt)^4 \tan^{-1}\left(\frac{yt}{xt}\right)$$

$$w(xt, yt) = t^4 \cdot x^4 \tan^{-1}\left(\frac{y}{x}\right) = t^4 w(x, y)$$

Thus, w is a homogeneous function of degree $n_2 = 4$.

Step 3: Set Up the Joint Operators for Both Components

The problem requires evaluating a combination of both the **First-Order Euler Theorem** and its **Second-Order Corollary**. Group the matching differential operators together for a general homogeneous function H of degree n:

$$\text{Total Operator Expression} = \left(x^2 \frac{\partial^2 H}{\partial x^2} + 2xy \frac{\partial^2 H}{\partial x \partial y} + y^2 \frac{\partial^2 H}{\partial y^2}\right) + \left(x \frac{\partial H}{\partial x} + y \frac{\partial H}{\partial y}\right)$$

Substitute the known theorems ($n(n-1)H$ and nH respectively) to create a simplified algebraic operator formula:

$$\text{Total Operator Expression} = n(n-1)H + nH$$

$$\text{Total Operator Expression} = (n^2 - n + n)H = n^2 H$$

Step 4: Apply the Unified Formula to v and w

Using the linearity property of partial derivatives, calculate the full expression value by applying our unified formula ($n^2 H$) to each component function independently:

For Component v ($n_1 = 3$):

$$\left(x^2 \frac{\partial^2 v}{\partial x^2} + 2xy \frac{\partial^2 v}{\partial x \partial y} + y^2 \frac{\partial^2 v}{\partial y^2}\right) + \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y}\right) = (3)^2 v = 9v$$

For Component w ($n_2 = 4$):

$$\left(x^2 \frac{\partial^2 w}{\partial x^2} + 2xy \frac{\partial^2 w}{\partial x \partial y} + y^2 \frac{\partial^2 w}{\partial y^2}\right) + \left(x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y}\right) = (4)^2 w = 16w$$

Summing both expressions yields the total value for u:

$$\text{Total Expression Value} = 9v + 16w$$

Substitute back the original definitions of v and w:

$$\text{Total Expression Value} = 9\left[x^3 \sin^{-1}\left(\frac{y}{x}\right)\right] + 16\left[x^4 \tan^{-1}\left(\frac{y}{x}\right)\right]$$

Step 5: Evaluate at the Given Boundary Conditions ($x = 1, y = 1$)

Substitute $x = 1$ and $y = 1$ directly into the consolidated expression:

$$\text{Total Value} = 9\left[(1)^3 \sin^{-1}\left(\frac{1}{1}\right)\right] + 16\left[(1)^4 \tan^{-1}\left(\frac{1}{1}\right)\right]$$

$$\text{Total Value} = 9\sin^{-1}(1) + 16\tan^{-1}(1)$$

Substitute the exact trigonometric values in radians ($\sin^{-1}(1) = \frac{\pi}{2}$ and $\tan^{-1}(1) = \frac{\pi}{4}$):

$$\text{Total Value} = 9\left(\frac{\pi}{2}\right) + 16\left(\frac{\pi}{4}\right)$$

$$\text{Total Value} = \frac{9\pi}{2} + 4\pi$$

Find a common denominator to combine the terms:

$$\text{Total Value} = \frac{9\pi + 8\pi}{2} = \frac{17\pi}{2}$$

Final Answer

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{17\pi}{2}$$